

Curriculum Vitae Caluã de Lacerda Pataca

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Lecturer in Computer Science at Birmingham City University, United Kingdom. Former Fulbright fellow and Ph.D. graduate from the Rochester Institute of Technology (RIT), researching the intersection of Human-Computer Interaction (HCI), Affective Computing, and Accessibility to design multimodal, adaptive, and user-centred captioning systems.

ACADEMIC & PROFESSIONAL EXPERIENCE

Lecturer in Computer Science	Birmingham City University	01/2026–Present
Graduate Research Assistant	Rochester Institute of Technology	08/2021–10/2025
Founder and Head of Design	Preface (Brazil)	10/2006–08/2021

EDUCATION

Rochester Institute of Technology (RIT), Rochester, New York, United States

Ph.D. in Computing and Information Sciences 08/2021–10/2025

Advisors: Dr. Matt Huenerfauth & Dr. Roshan Peiris

Thesis: Improving the Accessibility of Speech for Deaf and Hard-of-Hearing
Individuals Through Affective Captions

University of Campinas (Unicamp), Campinas, São Paulo, Brazil

M.Sc. in Electrical and Computer Engineering 08/2018–05/2021

Advisor: Dr. Paula Dornhofer Paro Costa

Thesis: Speech-Modulated Typography

B.A. and Licentiate degree in Visual Arts 02/2002–07/2007

Certificate programme in programming (technical high school from Unicamp) 02/1999–12/2002

PEER-REVIEWED CONFERENCE PAPERS

- [C12] IN PREPARATION Sunday David Ubur, **Caluã de Lacerda Pataca**, Sikiru Adewale, Seo-Yeon Kim, Joshua Butler, Brian Trager, and Denis Gracanin. "Speaker-Aware Captions for STEM Education: An Exploratory Study of Affect Cues and AI Clarification."

* Denotes equal contribution.

- [C11] Saad Hassan, Laleh Nourian, **Caluã de Lacerda Pataca**, Michelle M. Olson, Toni D'Aurio, Kanupriya Agarwal, Syeda Mah Noor Asad, Garreth W. Tigwell, and Matt Huenerfauth. ["ASL Educators' Perspectives on AI for Enhancing Student Learning in American Sign Language Education."](#) In *Proceedings of the 2026 ACM CHI Conference on Human Factors in Computing Systems*. 2026.
- [C10] **de Lacerda Pataca, Caluã***, Stephanie Patterson*, Roshan L. Peiris, and Matt Huenerfauth. ["Fuzzy Feelings: Arousal's Interpretive Noise and the Case for Acoustic-Based Haptics."](#) In *Proceedings of the 2026 ACM CHI Conference on Human Factors in Computing Systems*. 2026.
- [C9] **de Lacerda Pataca, Caluã***, SooYeon Ahn*, Suhyeon Yoo*, JooYeong Kim, Khai Truong, Jin-Hyuk Hong, Roshan Peiris, and Matt Huenerfauth. ["CuCap: Comparative Analysis of Customized Captioning for English and Korean d/Deaf and Hard-of-Hearing Users."](#) In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility*, pp 1-20. 2025.
- [C8] Jeremy Zhengqi Huang, **Caluã de Lacerda Pataca**, Liang-Yuan Wu, and Dhruv Jain. ["Cap-Tune: Adapting Non-Speech Captions With Anchored Generative Models"](#) In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility*, pp 1-18. 2025.
- [C7] **de Lacerda Pataca, Caluã**, Saad Hassan, Lloyd May, Michelle Olson, Toni D'Aurio, Roshan Peiris, and Matt Huenerfauth. ["Tactile Emotions: Multimodal Affective Captioning with Haptics Improves Narrative Engagement for d/Deaf and Hard-of-Hearing Viewers."](#) In *Proceedings of the 2025 ACM CHI Conference on Human Factors in Computing Systems*, pp. 1-17. 2025.
- [C6]  **de Lacerda Pataca, Caluã**, Saad Hassan, Nathan Tinker, Roshan Peiris, and Matt Huenerfauth. ["Caption Royale: Exploring the Design Space of Affective Captions from the Perspective of Deaf and Hard-of-Hearing Individuals."](#) In *Proceedings of the 2024 ACM CHI Conference on Human Factors in Computing Systems*, pp. 1-17. 2024. Honourable mention.
- [C5]  Hassan, Saad, **Caluã de Lacerda Pataca**, Laleh Nourian, Garreth Tigwell, Briana Davis, and Will Wagman. ["Designing and Evaluating an Advanced Dance Video Comprehension Tool with In-situ Move Identification Capabilities."](#) In *Proceedings of the 2024 ACM CHI Conference on Human Factors in Computing Systems*, pp. 1-19. 2024. Honourable mention.
- [C4] **de Lacerda Pataca, Caluã**, Matthew Watkins, Roshan Peiris, Sooyeon Lee, and Matt Huenerfauth. ["Visualization of Speech Prosody and Emotion in Captions: Accessibility for Deaf and Hard-of-Hearing Users."](#) In *Proceedings of the 2023 ACM CHI Conference on Human Factors in Computing Systems*, pp. 1-15. 2023.

- [C3] Hassan, Saad*, Akhter Al Amin*, **Caluã de Lacerda Pataca***, Diego Navarro, Alexis Gordon,  Sooyeon Lee, and Matt Huenerfauth. ["Support in the Moment: Benefits and Use of Video-Span Selection and Search for Sign Language Video Comprehension among ASL Learners."](#) In *Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility*, pp. 1–14. 2022. Best paper nomination.
- [C2] **de Lacerda Pataca, Caluã**, and Paula Dornhofer Paro Costa ["Speech modulated typography: towards an affective representation model."](#) In *Proceedings of the 25th International Conference on Intelligent User Interfaces*, pp. 139–143. 2020.
- [C1] **Caluã de Lacerda Pataca**, and Paula Dornhofer Paro Costa ["Tipografia modulada pela fala: avaliação de um algoritmo de geração de prosódia visual em textos."](#) In *Blucher Design Proceedings* 6, no. 4. 2019.

JOURNAL ARTICLES

- [J2] Hassan, Saad, **Caluã de Lacerda Pataca***, Laleh Nourian*, Akhter Al Amin*, Diego Navarro, Sooyeon Lee, Alexis Gordon, Matthew Watkins, Garreth Tigwell, and Matt Huenerfauth. ["Exploring the Benefits and Applications of Video-Span Selection and Search for Real-Time Support in Sign Language Video Comprehension among ASL Learners."](#) *ACM Transactions on Accessible Computing* 17, no. 3 (2024): 1–35.
- [J1] **de Lacerda Pataca, Caluã**, and Paula Dornhofer Paro Costa. ["Hidden Bawls, Whispers, and Yelps: Can Text Convey the Sound of Speech, Beyond Words?"](#) *IEEE Transactions on Affective Computing* 14, no. 1 (2022): 6–16.

PATENTS

- Paula Dornhofer Paro Costa, and **Caluã de Lacerda Pataca**. Método para composição de closed-captions com parâmetros tipográficos representando visualmente qualidades acústicas da fala (Pending. Process BR-10/2021-016125-6).
- Ricardo Alexandre Da Silva Shimabukuro Victorio, Gustavo Vilela Vargas, Fernando Augusto Chufi, Thiago Yoshihiro Ito Takahashi, Fábio San Juan, **Caluã de Lacerda Pataca**, José Luis Ponciano Bomfim, Adriana Ramos, Guilherme de Oliveira Feliciano, and Maria Cristina Marcucci Ribeiro (2012). Traceability system applied to farm production activities, industrialization and commercialization of bee products. U.S. Patent Application No. 12/596,612.

INVITED TALKS & GUEST LECTURES

- 02/26 Research on Deaf Community Preferences for Expressive Captions, presented with Matt Huenerfauth · Expressive Captioning Summit convened by the Academy of Motion Picture Arts and Sciences's Science and Technology Council
- 09/25 Adding Nuance: Emotion-Aware, Flexible Captions for Speech Access · State of the Science Conference at Gallaudet University
- 02/25 Improving Access, Satisfaction, and Engagement with Video through Affective Captions · William & Mary Department of Computer Science
- 10/24 Experimental Research Design · HCIN-600 Research Methods, RIT
- 09/24 Captions beyond words: Paralinguistic accessibility for Deaf and Hard-of-Hearing individuals · 'Artificial Intelligence for Multimodal Signal Processing' Lab at the University of Campinas School of Electrical and Computer Engineering
- 03/24 Captions beyond words: Paralinguistic accessibility for Deaf and Hard-of-Hearing individuals · Department of Access Services, RIT
- 11/23 Enhancing Affective Captions with Haptic Feedback · Haptics for Inclusion Symposium, University of Borås, Sweden
- 10/23 Captions beyond words: Paralinguistic accessibility for d/Deaf and Hard-of-Hearing individuals · Comm. Sciences and Disorders Department, University of Houston
- 03/23 Visualization of Speech Prosody and Emotion in Captions: Accessibility for Deaf and Hard-of-Hearing Users · ISTE-266 Design for Accessibility, RIT
- 10/22 Introduction to experimental design · HCIN-600 Research Methods, RIT
- 09/22 Identifying emotions, moods, and emphasis through captions · Readability Consortium monthly meeting
- 06/13 Design Issues in Web Application and Website Projects · Semana de Atualização em Informática, UTFPR (Ponta Grossa), Brazil, presented together with Lucas V. Reis

TEACHING

- **Qualitative Research Methods at the Ph.D. PG Cert course** April and June 2026
Introductory lecture for the February and May Ph.D. intakes
- **CMP7158 Research Methods and Project Management** 2026
Prepared and delivered half of the module's lectures, as well as all activities and quizzes, and completed the associated marking.
- **Teaching Assistant for the ISTE-798 Future Interactions course** 2025
Dr. Garreth Tigwell's class at the Rochester Institute of Technology

- **Experimental Research Design Lecture**

2022 & 2024

HCIN-600 Research Methods class under Dr. Roshan Peiris at the Rochester Institute of Technology

MENTORSHIP & SUPERVISION

Student Supervision and Mentoring

Birmingham City University – Dissertation and Project Supervision (2026–)

- *Lucy Ighometuwho Israel; Zainab Sajid; Ponthakorn Methavichit; Myah Majid; Usman Ellahi; Raja Asad Khalil; Muhammad Atif* (MSc): Dissertation supervision.
- *Panu Khumchoo-Joonpijarn; James Taylor; Onaopemiposi Adeyeye; Munib Sarfraz* (BSc): Final-year project co-supervision with Dr Sima Iranmanesh.

Rochester Institute of Technology – Research Supervision and Mentoring (2022–2025)

- *Stephanie Patterson* (MSc HCI, 2025–): Data collection for two of my studies; advising independent study; co-author on pending and planned submissions.
- *Sidney Grabosky* (MSc HCI, 2025–): Game design and study; co-author on upcoming paper submission.
- *Matthew Watkins; Michelle Olson* (MSc HCI, 2022; 2024): Study coordination and participant recruitment across multiple lab and remote sessions.
- *Yiyuan (Steve) Wufeng* (BS CS, 2023–24): Mentored through internship and co-op; advised student-led study using our jointly developed captioning pipeline.
- *Andrew Chacon* (BS CE, 2023): Supervised development of caption-rendering pipeline for my research programme.
- *Russell Lee* (BS HCC, 2023): Supervised software projects including an open-source jsPsych plugin.
- *Nathan Tinker; Toni D'Aurio* (BS ASL-English Interpretation, 2022; 2024): Coordinated ASL video recording for dictionary project; supervised recruitment and data collection.

SCHOLARSHIPS & AWARDS

- 2021–2026 Fulbright/Capes Full Doctorate Program · *The highly competitive Fulbright Scholarship programme supports doctoral study at top U.S. institutions. (\$120,000)*
- Two honourable mentions at CHI'24 · *Top 5% of conference papers.*
- Best paper nomination at ASSETS'22 · *Top 6% of conference papers.*

- 2022 CRA-WP Grad Cohort Workshop for IDEALS · *This competitive programme for students underrepresented in the computing field provides mentoring for graduate students across North America.*
- SIGCHI Student Travel Grant (SSTG) to present at the 2020 International Conference on Intelligent User Interfaces at Cagliari, Italy · *Competitive programme providing support to students attending SIGCHI-sponsored or co-sponsored conferences.*
- 3 bronze medals, 1 silver medal, and 2 Popular Vote Trophies in 2018, 2019, and 2020 editions of the Brasil Design Award · *The awards were presented at the annual ceremony hosted by the Association for Brazilian Design Companies (ABEDESIGN).*
- Selected project for the 13th Brazilian Graphic Design Biennial · *Prestigious juried exhibition organised by the Brazilian Association of Graphic Designers.*

PROFESSIONAL SERVICE

Program Committee Member, Technical Papers 2026
ACM SIGACCESS Conference on Computers and Accessibility (ASSETS'26)

Associate Chair, Posters 2026
ACM Conference on Human Factors in Computing Systems (CHI'26)

Associate Chair, Late-Breaking Work 2025
ACM Conference on Human Factors in Computing Systems (CHI'25)

Web and Graphic Design Chair 2024–25
ACM SIGACCESS Conference on Computers and Accessibility (ASSETS'25)

Reviewer 2026
Interacting with Computers

Technical Paper Reviewer 2026
IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR'26)

Reviewer 2026
IEEE Computer Magazine

Technical Paper Reviewer 2023–26
ACM Conference on Human Factors in Computing Systems (CHI)

Late-Breaking Work Reviewer 2023
ACM Conference on Human Factors in Computing Systems (CHI'23)

Communications Chair 2022–24
Center for Accessibility and Inclusion Research (CAIR Lab), Rochester Institute of Technology

Technical Paper Reviewer

2021

Congresso Nacional de Iniciação Científica em Design (CONGIC)

PROFESSIONAL DEVELOPMENT

- **IwC: Reviewing Academy**, with Carine Lallemand & Helen Petrie 2025
- **The Expressive Function of Prosody in Speech**, with Plínio A. Barbosa 2019
- **Scientific Writing Workshop**, with Inês Signorini 2019
- **Introduction to Computational Design**, with Guilherme Vieira 2017
- **Graphic Production**, with Aline Valli 2013
- **Branding and Culture**, with André Stolarski 2012

POSTER PRESENTATIONS

- [P3] **Caluã de Lacerda Pataca**, Matthew Watkins, Sooyeon Lee, and Matt Huenerfauth. (2022) Developing a captioning style that can help Deaf and Hard-of-hearing individuals identify a speaker's emotions, moods, and emphasis. *AI@RIT*
- [P2] **Caluã de Lacerda Pataca**, and Matt Huenerfauth. (2022) Visualizing paralinguistic dimensions of speech through typography. *2022 CRA - WP Grad Cohort Workshop for IDEALS*
- [P1] **Caluã de Lacerda Pataca**, and Paula Dornhofer Paro Costa. (2019) Tipografia modulada pela fala. In: *Proceedings of the XII EADCA Encontro de Alunos e Docentes do DCA*.

EXTERNAL RESEARCH COLLABORATIONS

Published and forthcoming collaborations Stanford University (US); University of Michigan (US); University of Toronto (Canada); Gwangju Institute of Science and Technology (South Korea).

Active and developing collaborations Rochester Institute of Technology (US); University of Toronto (Canada); Monash University (Australia); Tulane University (US); Virginia Tech (US); Bournemouth University (UK); Universidade Federal de Minas Gerais (Brazil); University of Birmingham (UK); University of Campinas (Brazil).

REFERENCES

Matt Huenerfauth, Dean and Professor at the Rochester Institute of Technology, USA

Ph.D. co-advisor · matt.huenerfauth@rit.edu

Roshan Peiris, Assistant Professor at the Rochester Institute of Technology, USA

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Paula Dornhofer Paro Costa, Professor and Research Scientist at the University of Campinas, Brazil

M.Sc. advisor · paulad@unicamp.br

Luís Cláudio Portugal do Nascimento, Professor at the University of São Paulo, Brazil

BA professor · claudioportugal@usp.br